Pre-existing phytoplankton biomass concentrations shape coastal plankton response to fire-generated ash leachate

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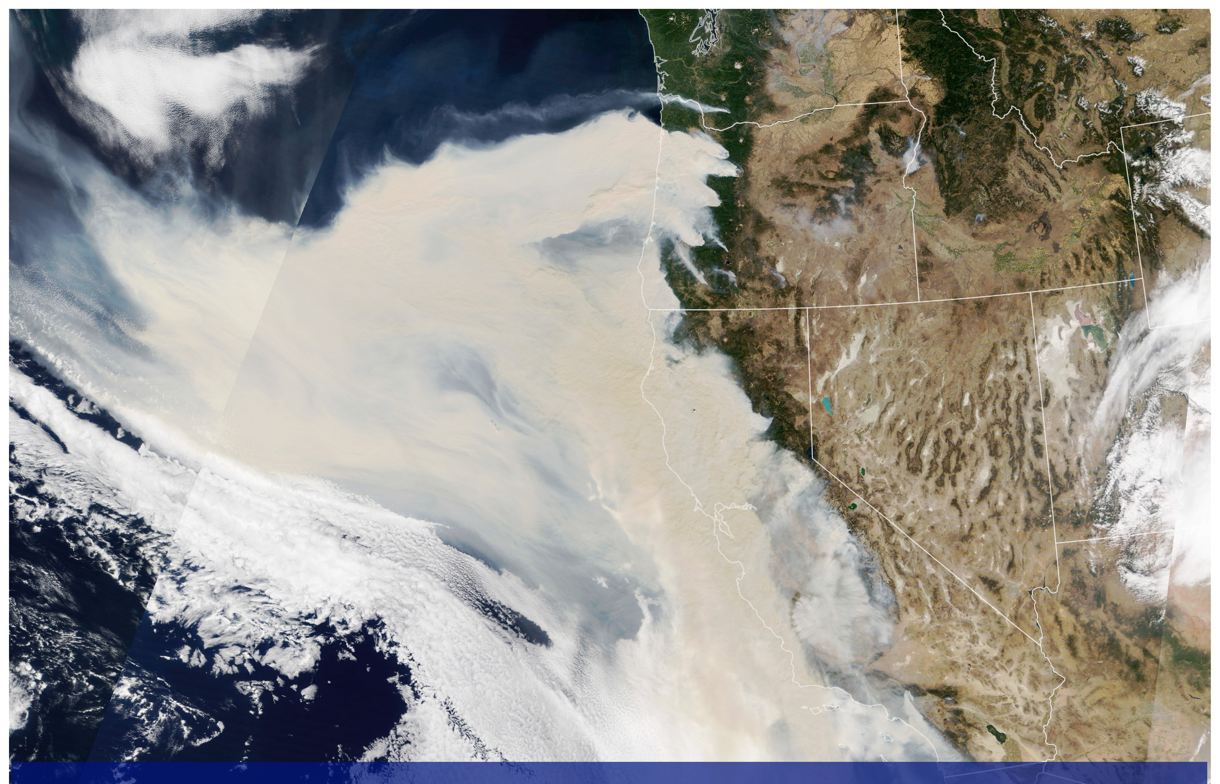
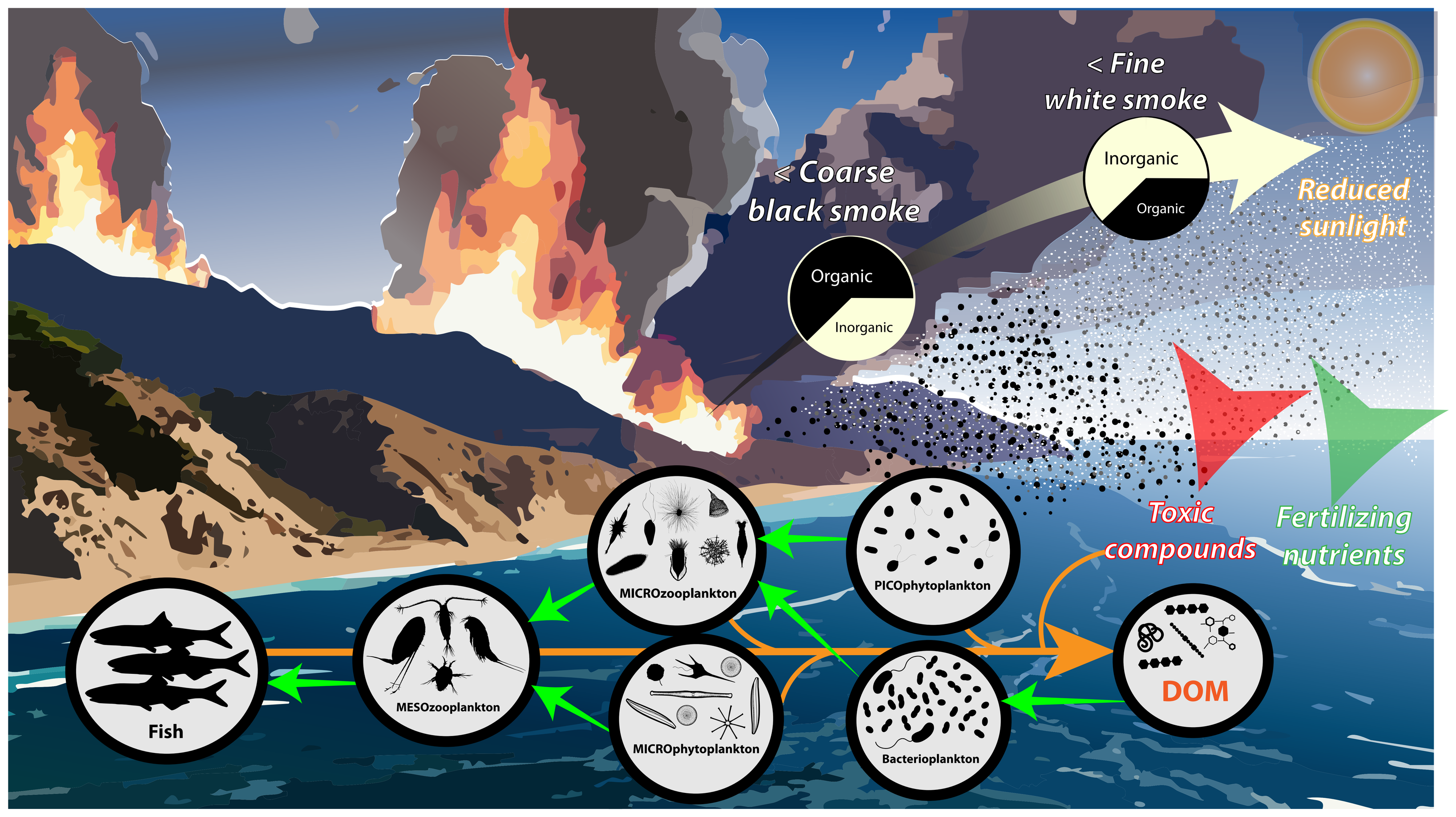


MODIS-Aqua image of smoke transport from 2020 Labor Day fires on U.S. West Coast. *Credit: NASA Earth Observatory*



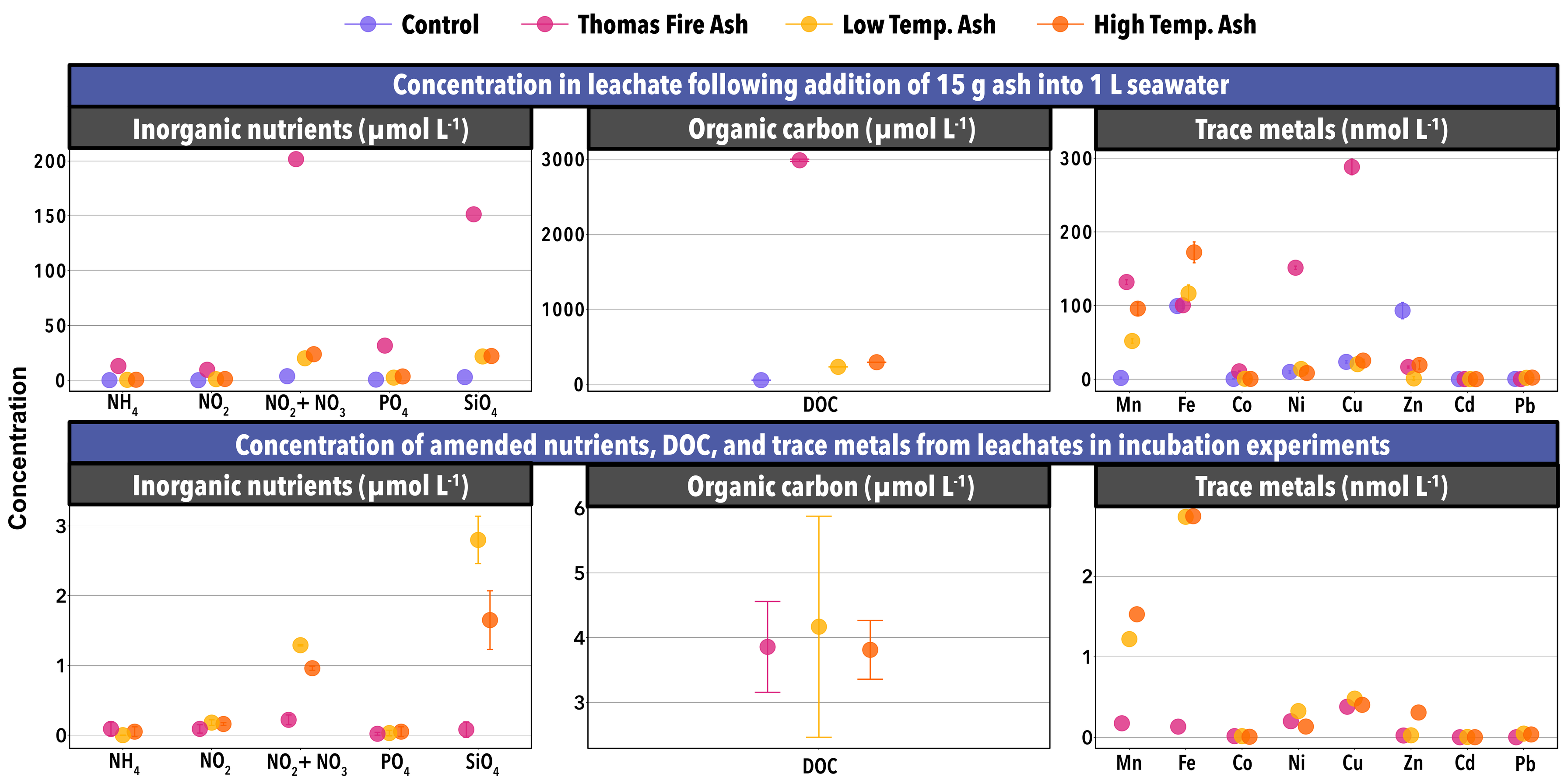
Pathways of wildfire smoke: coarse smoke (i.e., ash) from coastal wildfires, rich in organic matter and low in minerals, is likely to settle near the fire source. Fine smoke, with lower organic content and higher mineral composition, disperses farther. **Wildfire smoke deposition can introduce both fertilizing nutrients**, such as inorganic nitrogen and iron, **and toxic compounds**, including dissolved organic matter (DOM) species like aromatic hydrocarbons, affecting marine trophic levels. Additionally, wildfire smoke on the ocean surface can alter sunlight penetration, impacting photosynthesis.

How does the material leached from different ash types affect marine plankton across a range of environmental conditions?

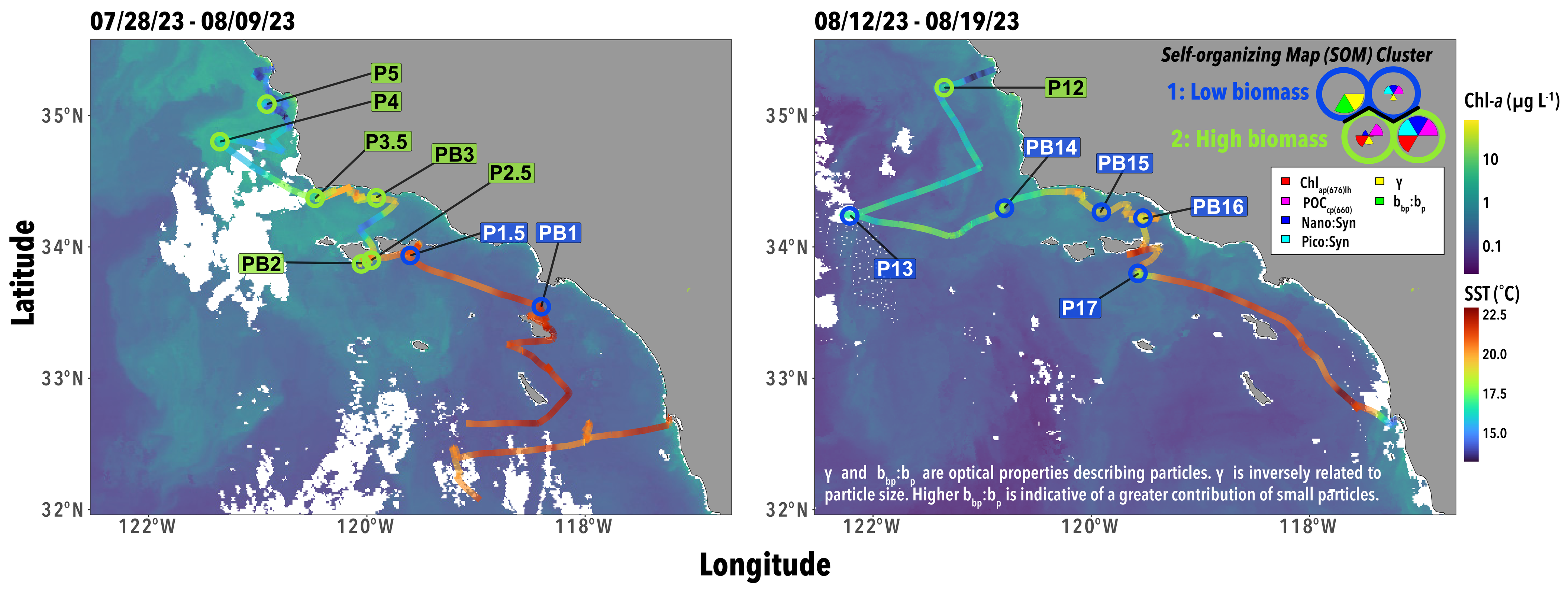
Take homes:

The pre-existing surface ocean phytoplankton biomass conditions determine the fate of ash leachate through plankton.
Ash leachate DOM has a greater potential to accumulate in high phytoplankton biomass water.
Phytoplankton accumulation occurs following the addition of ash leachate due to the decoupling of phytoplankton division and microzooplankton grazing.

1 Ash leachate composition depends on source material and fire conditions.



2 Ash leachates were tested in bacterial DOC remineralization experiments (5 d) and phytoplankton dilution experiments (24 h) in non-smoke conditions along the California coast during the summer, at 14 stations with low and high phytoplankton biomass.

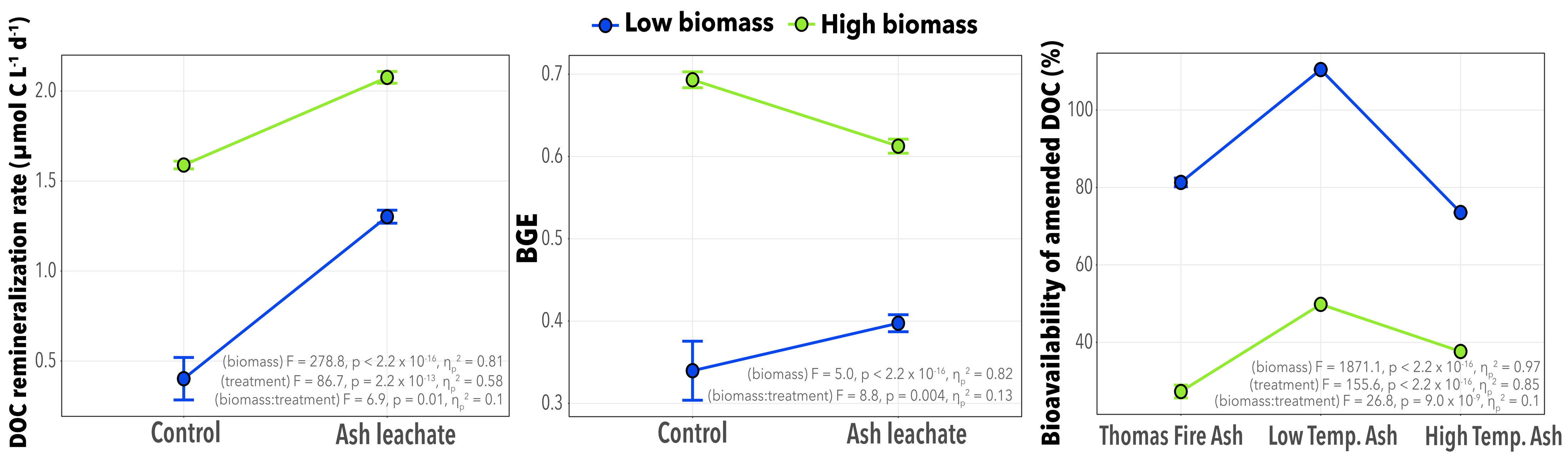


3 DOM composition in the environment prior to ash deposition likely dictates the transfer of ash-derived DOM from bacteria to higher trophic levels.

(A) DOM remineraliztion rates are enhanced with the addition of ash leachate.

(B) Bacterial growth efficiency changes with the addition of ash leachate depending on the biomass condition.

(C) Short-term bioavailability of the ash leachate DOM in high biomass water is lower than in low biomass water. Ash-derived DOM has a stronger potential to accumulate in high biomass water.

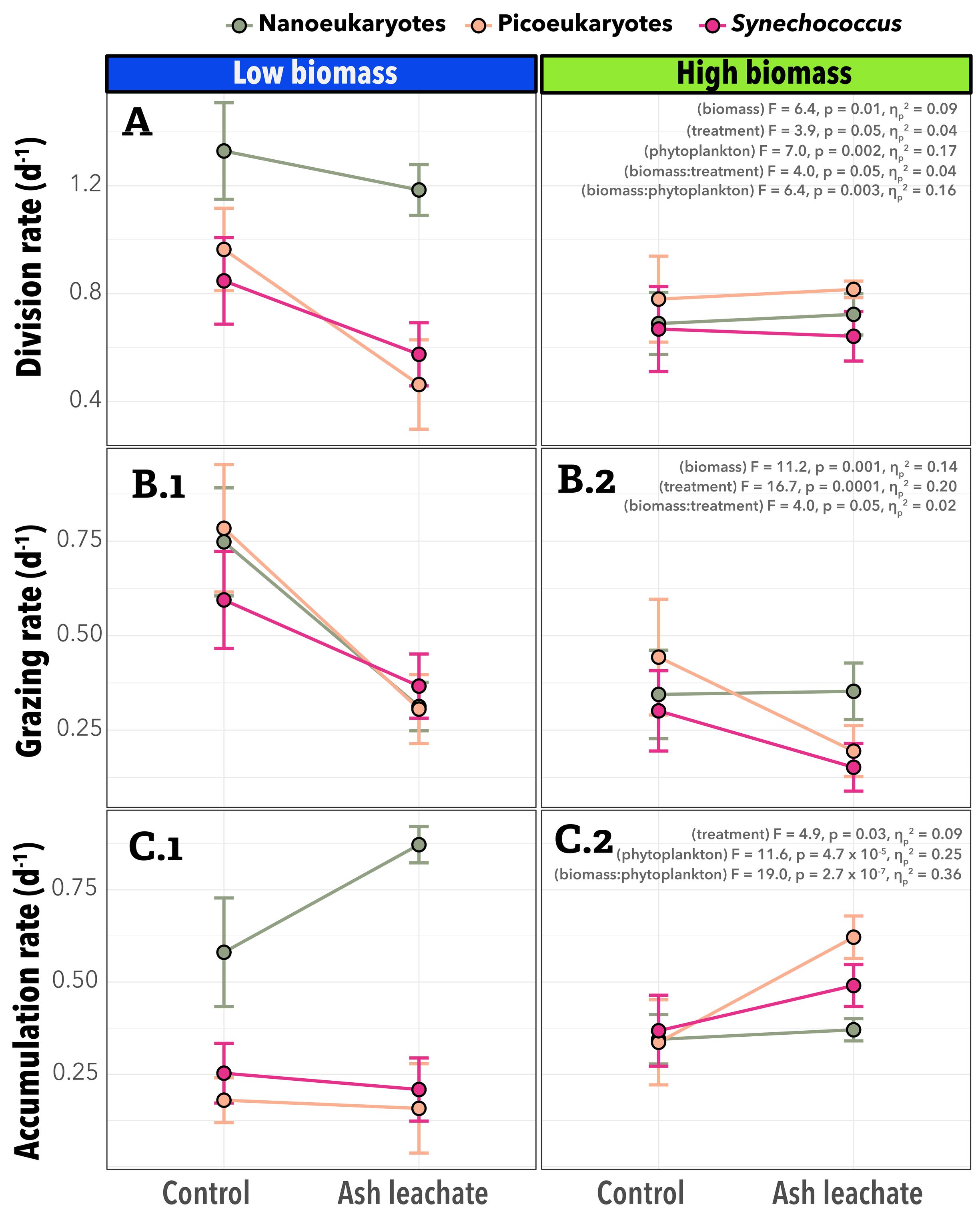


4 The influence of ash leachate on phytoplankton accumulation depends on pre-existing biomass levels.

(A) The addition of the ash leachates had a neutral to negative effect on phytoplankton division rates. Toxicity may have outweighed fertilizing characteristics of the ash leachate in the short-term.

(B.1) Decreased microzooplankton grazing on all phytoplankton increased the **(C.1)** accumulation of nanoeukaryotes in low-biomass water.

(B.2) Decreased microzooplankton grazing of smaller phytoplankton may increase the **(C.2)** accumulation of cyanobacteria and picoeukaryotes in high-biomass water.



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